

Linear MDPs

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Outline

- 1 Motivation & Setting
- 2 The Linear MDP Model
- 3 Bridge: Bellman Rank & Generalization

Recap: From Bandits to MDPs

- **Tabular MDPs (UCBVI):** We achieved $\tilde{O}(\text{poly}(H)\sqrt{SAK})$ regret.
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 - Great, but scales poorly when the state space $\mathcal{S}$ is enormous.
- **Linear Bandits (LinUCB):** We achieved $\tilde{O}(d\sqrt{K})$ regret.
 - No dependence on the number of arms! We exploit the feature dimension d .
- **Today's Goal:** Can we combine these ideas?
 - We want an algorithm for episodic MDPs with $\text{poly}(d, H, \sqrt{K})$ regret.
 - **Crucially:** Zero explicit dependence on $|\mathcal{S}|$ or $|\mathcal{A}|$.

Episodic MDP Setting

- **Horizon:** H steps per episode, total K episodes.
- **State/Action:** Large state space $\mathcal{S}$, action space $\mathcal{A}$, fixed start s_0 .
- **Rewards:** Deterministic, known, and bounded: $r_h(s, a) \in [0, 1]$.
 - *Note: Unknown rewards just require an extra Hoeffding bonus. We assume them known to isolate the difficulty of learning transitions.*

The Interaction Loop: In episode k , the learner chooses a policy $\pi^k = \{\pi_h^k\}_{h=0}^{H-1}$, and generates a trajectory:

$$s_0^k = s_0, \quad a_h^k = \pi_h^k(s_h^k), \quad s_{h+1}^k \sim P_h^*(\cdot \mid s_h^k, a_h^k)$$

Regret:

$$R_K := \sum_{k=0}^{K-1} (V^* - V^{\pi^k})$$

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The Linear Transition Model

We assume the transition kernel is *linear in known features*.

Assumption (Linear MDP)

There exists a known feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ with $\sup_{s,a} \|\phi(s,a)\|_2 \leq 1$, and unknown parameters $\{\mu_h^*\}_{h=0}^{H-1}$ with $\mu_h^* \in \mathbb{R}^{|\mathcal{S}| \times d}$ such that for all h and (s,a) ,

$$P_h^*(\cdot \mid s, a) = \mu_h^* \phi(s, a).$$

Equivalently, for each $s' \in \mathcal{S}$, $P_h^*(s' \mid s, a) = \mu_h^*(s')^\top \phi(s, a)$.

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Low-rank view (matrix form). View P_h^* as a matrix in $\mathbb{R}^{(SA) \times S}$ and $\Phi \in \mathbb{R}^{(SA) \times d}$. Then

$$P_h^* = \Phi (\mu_h^*)^\top, \quad \text{so} \quad \text{rank}(P_h^*) \leq d.$$

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Normalization (used to bound coefficients). For all $f : \mathcal{S} \rightarrow \mathbb{R}$,

$$\|(\mu_h^*)^\top f\|_2 \leq \sqrt{d} \|f\|_\infty.$$

Examples & Intuition: What is a Linear MDP?

Warm-up (tabular): take $\phi(s, a)$ one-hot in $\mathbb{R}^{SA}$, so $d = SA$.

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Example 1: Latent mixture / “topic” model

- Known mixture weights: $\phi(s, a) \in \Delta(d)$.
- Unknown components: $\mu_{h,1}^*, \dots, \mu_{h,d}^* \in \Delta(\mathcal{S})$.
- Mixture of d base transition kernels:

$$P_h^*(\cdot \mid s, a) = \sum_{i=1}^d \phi_i(s, a) \mu_{h,i}^*(\cdot).$$

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Example 2: Known state embedding + unknown linear latent dynamics

- Suppose we are given a *state embedding* $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ (learned or engineered).
- Unknown dynamics act linearly in the embedding space: $W_h^* \in \mathbb{R}^{d \times d}$,

$$P_h^*(s' \mid s, a) = \psi(s')^\top W_h^* \phi(s, a) \quad (\text{with normalization so this is a distribution}).$$

- Interpretation: next-state probabilities depend on (s, a) only through $\phi(s, a)$, and the dependence on s' is summarized by $\psi(s')$.

The “Magic” Property: Linear Backups

Key consequence: for any bounded $f : \mathcal{S} \rightarrow \mathbb{R}$, $(P_h^* f)(s, a)$ is linear in $\phi(s, a)$.

Lemma (Linearization of $P_h^* f$)

Fix h and bounded f . Let $w_{h,f}^* := (\mu_h^*)^\top f \in \mathbb{R}^d$. Then

$$(P_h^* f)(s, a) := \mathbb{E}_{s' \sim P_h^*(\cdot | s, a)}[f(s')] = \phi(s, a)^\top w_{h,f}^*.$$

If $\|f\|_\infty \leq H$, then $\|w_{h,f}^*\|_2 \leq H\sqrt{d}$.

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Proof sketch:

$$\begin{aligned}(P_h^* f)(s, a) &= \sum_{s'} f(s') P_h^*(s'|s, a) = \sum_{s'} f(s') \mu_h^*(s')^\top \phi(s, a), \\ &= \left(\sum_{s'} f(s') \mu_h^*(s') \right)^\top \phi(s, a) = ((\mu_h^*)^\top f)^\top \phi(s, a).\end{aligned}$$

Implication: A Strong Form of Completeness

Recall (Bellman completeness): $f \in \mathcal{F} \Rightarrow \mathcal{T}_h f \in \mathcal{F}$ (closure only for f in the hypothesis class).

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Linear MDP is stronger (for the transition part): for each stage h ,

$$\forall f : \mathcal{S} \rightarrow \mathbb{R} \text{ bounded,} \quad (P_h^* f)(s, a) = \mathbb{E}[f(s') \mid s, a] \in \text{span}(\phi).$$

Equivalently, there exists $w_{h,f}^* \in \mathbb{R}^d$ such that

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Why this matters for DP / least squares:

- V_{h+1}^* is nonlinear (max over actions), and $\widehat{V}_{h+1}^k$ is even messier (bonus + clipping).
- **Still:** the *regression target* $(s, a) \mapsto (P_h^* \widehat{V}_{h+1}^k)(s, a)$ is always exactly linear in $\phi(s, a)$.
- So each stage reduces to a *well-specified* linear regression problem for $w_{h, \widehat{V}_{h+1}^k}^*$.

Planning with a Known Model

Suppose we knew the transition parameters $\{\mu_h^*\}_{h=0}^{H-1}$.

Backward DP: $V_H \equiv 0$, and for $h = H - 1, \dots, 0$,

$$Q_h^*(s, a) = r_h(s, a) + (P_h^* V_{h+1}^*)(s, a), \quad V_h^*(s) = \max_{a \in \mathcal{A}} Q_h^*(s, a).$$

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Linear MDP plug-in: for any f ,

$$(P_h^* f)(s, a) = \phi(s, a)^\top (\mu_h^*)^\top f.$$

So the Bellman backup can be written as

$$Q_h^*(s, a) = r_h(s, a) + \phi(s, a)^\top w_h^*, \quad w_h^* := (\mu_h^*)^\top V_{h+1}^* \in \mathbb{R}^d.$$

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Key takeaway: we never need the full $P_h^*(\cdot \mid s, a)$. To plan, we can estimate the *linear functional*:

$$(s, a) \mapsto (P_h^* V_{h+1}^*)(s, a) = \phi(s, a)^\top w_{h, V_{h+1}^*}^*.$$

Lin-UCBVI does this with ridge regression + an elliptical bonus.

Stage-wise Ridge Regression (Estimating $P_h^* f$)

At episode k , stage h , we have past transitions $D_h^k = \{(s_h^i, a_h^i, s_{h+1}^i)\}_{i < k}$ and features $x_h^i := \phi(s_h^i, a_h^i) \in \mathbb{R}^d$.

Design matrix:

$$\Lambda_h^k := \lambda I + \sum_{i=0}^{k-1} x_h^i (x_h^i)^\top.$$

Ridge fit for a given target function $f : \mathcal{S} \rightarrow \mathbb{R}$:

$$\hat{w}_{h,f}^k := (\Lambda_h^k)^{-1} \sum_{i=0}^{k-1} x_h^i f(s_{h+1}^i), \quad (\hat{P}_h^k f)(s, a) := \phi(s, a)^\top \hat{w}_{h,f}^k.$$

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Interpretation: we never estimate $P_h^*(\cdot \mid s, a)$; we only estimate the *linear functional* $(s, a) \mapsto (P_h^* f)(s, a)$ for the specific f used in planning (namely $f = \hat{V}_{h+1}^k$).

UCB Bonus: Uncertainty in Direction $\phi(s, a)$

Under the linear MDP model, for each fixed f there is a true coefficient $w_{h,f}^* = (\mu_h^*)^\top f$ with $(P_h^* f)(s, a) = \phi(s, a)^\top w_{h,f}^*$. Ridge gives $\hat{w}_{h,f}^k$.

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High-probability form (LinUCB-style): for all (s, a) ,

$$|\phi(s, a)^\top (\hat{w}_{h,f}^k - w_{h,f}^*)| \leq \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

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$$|\phi(s, a)^\top (\hat{w}_{h,f}^k - w_{h,f}^*)| \leq \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

So we use the exploration bonus:

$$b_h^k(s, a) := \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

This is large when $\phi(s, a)$ lies in a poorly-sampled direction (small eigenvalues of Λ_h^k), and shrinks as we collect more data.

Optimistic Planning (Lin-UCBVI)

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Execute π^k for one episode, then update $\Lambda_h^{k+1} = \Lambda_h^k + \phi(s_h^k, a_h^k) \phi(s_h^k, a_h^k)^\top$.

The Regret Guarantee

Theorem (Regret of Lin-UCBVI)

Assume the Linear MDP model and $r_h(s, a) \in [0, 1]$. Run Lin-UCBVI with $\lambda \geq 1$ and a bonus scale $\beta = \tilde{O}(Hd)$ (from a uniform confidence bound).

Then with probability at least $1 - \delta$, the cumulative regret satisfies:

$$R_K = \sum_{k=0}^{K-1} (V^* - V^{\pi^k}) \leq c \beta H \sqrt{K d \log\left(1 + \frac{K}{\lambda}\right)} + \mathcal{O}\left(H \sqrt{K \log(1/\delta)}\right),$$

for a universal constant c .

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for a universal constant c .

Takeaways:

- The $\mathcal{O}(\sqrt{K})$ martingale noise term is dominated by the primary β term.
- Absorbing lower-order terms gives: $R_K = \tilde{O}\left(H^2 \sqrt{d^3 K}\right)$.
- No explicit dependence on $|\mathcal{S}|$ or $|\mathcal{A}|$ (structure captured entirely by d).

Proof Roadmap (What are the three moving parts?)

We reuse the same three ideas as UCBVI + LinUCB:

- ① **(Confidence)** Uniformly control value-dependent regression targets:

$$|(\hat{P}_h^k f)(s, a) - (P_h^* f)(s, a)| \leq \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}} \quad \text{for } f = \hat{V}_{h+1}^k.$$

- ② **(Optimism)** Backward induction shows $\hat{V}_h^k(s) \geq V_h^*(s)$ for all k, h, s .

- ③ **(Simulation + summation)** For each episode k ,

$$V^* - V^{\pi^k} \leq 2 \sum_{h=0}^{H-1} \mathbb{E}_{d_h^{\pi^k}} [b_h^k(s_h, a_h)],$$

then sum using an **elliptical potential** (log-det) argument.

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then sum using an **elliptical potential** (log-det) argument.

Today's only new wrinkle vs LinUCB: the target $f = \hat{V}_{h+1}^k$ is random, so we need *uniform* confidence over the whole optimistic value-function class.

The Statistical Hurdle: Why Uniform Confidence?

In LinUCB we estimate a fixed parameter μ^* . Here we estimate a *value-dependent* coefficient:

$$\hat{w}_{h, \hat{V}_{h+1}^k}^k = (\Lambda_h^k)^{-1} \sum_{i=0}^{k-1} \phi(s_h^i, a_h^i) \hat{V}_{h+1}^k(s_{h+1}^i).$$

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The catch:

- $\hat{V}_{h+1}^k$ is a **random function** built from past data (and bonuses).
- So the “labels” $\hat{V}_{h+1}^k(s_{h+1}^i)$ are correlated with the design/history.
- A fixed-function concentration bound is not enough; we need a bound that holds *simultaneously for all optimistic value functions the algorithm may produce*.

Step (a): Uniform Confidence over Optimistic Values

Proposition (Uniform model confidence (informal))

With probability at least $1 - \delta$, simultaneously for all episodes k , stages h , all (s, a) , and the specific $f = \hat{V}_{h+1}^k$ produced by the algorithm,

$$|(\hat{P}_h^k f)(s, a) - (P_h^* f)(s, a)| \leq \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

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Where does β come from? Self-normalized martingale concentration + an ϵ -net over the optimistic value class (max + bonus + clipping), which yields $\beta = \tilde{O}(Hd)$.

Step (b): Optimism (Backward Induction)

On the uniform confidence event, for every episode k , stage h , and state s ,

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Reason (one line): for any (s, a) ,

$$(P_h^* \widehat{V}_{h+1}^k)(s, a) \leq (\widehat{P}_h^k \widehat{V}_{h+1}^k)(s, a) + b_h^k(s, a),$$

so $r_h + P_h^* V_{h+1}^* \leq r_h + b_h^k + \widehat{P}_h^k \widehat{V}_{h+1}^k$ (and clipping preserves the inequality).

Step (c): Simulation Lemma (Per-episode bound)

Let $d_h^{\pi^k}(s, a) = \Pr(s_h = s, a_h = a \mid s_0, \pi^k, P^*)$ be the stage- h occupancy.

On the uniform confidence event, for every episode k ,

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Plug in the LinUCB-style bonus:

$$b_h^k(s, a) = \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}} \implies V^* - V^{\pi^k} \leq 2\beta \sum_{h=0}^{H-1} \mathbb{E}_{d_h^{\pi^k}} [\|\phi(s_h, a_h)\|_{(\Lambda_h^k)^{-1}}].$$

Summation: Elliptical Potential (Log-det)

Summing the simulation bound over episodes reduces regret to

$$R_K \leq 2\beta \sum_{h=0}^{H-1} \sum_{k=0}^{K-1} \|\phi(s_h^k, a_h^k)\|_{(\Lambda_h^k)^{-1}}.$$

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Fix a stage h and write $x_k := \phi(s_h^k, a_h^k)$, $\Lambda_{k+1} = \Lambda_k + x_k x_k^\top$. Matrix determinant lemma gives the standard potential bound:

$$\sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}}^2 \leq 2 \log \frac{\det(\Lambda_K)}{\det(\Lambda_0)} \leq 2d \log \left(1 + \frac{K}{\lambda}\right).$$

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By Cauchy–Schwarz,

$$\sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}} \leq \sqrt{K \sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}}^2} \leq \sqrt{2K d \log \left(1 + \frac{K}{\lambda}\right)}.$$

Sum over h and multiply by β to get the theorem.

The Question: What makes RL Generalize?

We have seen two successful "recipes" for avoiding $|\mathcal{S}|$ -dependence:

- 1 **Tabular UCBVI**: Estimate the full local transition.
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The General Problem: What if we don't know the features $\phi(s, a)$? Or what if the transitions are non-linear, but some other structural bottleneck exists?

The Goal: We seek a **structural complexity measure** that tells us when trajectory data collected under policy f can effectively "test" the quality of another hypothesis g .

The Core Object: Average Bellman Error

Define the one-step Bellman residual of a hypothesis $g \in \mathcal{H}$ as:

$$\ell_h(s, a, s'; g) := Q_{h,g}(s, a) - r_h(s, a) - V_{h+1,g}(s')$$

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We measure the quality of g using its **Average Bellman Error** under the roll-in distribution induced by the policy of f , denoted $d_h^{\pi_f}$:

$$\mathcal{E}_h^Q(f, g) := \mathbb{E}_{(s,a) \sim d_h^{\pi_f}, s' \sim P_h^*} [\ell_h(s, a, s'; g)]$$

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Key Intuition:

- If $g = f^*$ (realizability), then $\mathcal{E}_h(f, f^*) = 0$ for every roll-in policy f .
- Finding a near-optimal policy is equivalent to finding a g that has low Bellman error under its own distribution ($\mathcal{E}_h(g, g) \approx 0$).

Defining Bellman Rank

Consider a massive matrix M_h where rows are "Roll-in Policies" $f \in \mathcal{H}$ and columns are "Tested Hypotheses" $g \in \mathcal{H}$. Each entry is the cross-error $\mathcal{E}_h(f, g)$.

Definition (Q -Bellman Rank)

The pair $(\mathcal{M}, \mathcal{H})$ has Q -Bellman rank at most d if there exist maps $X_h, W_h : \mathcal{H} \rightarrow \mathbb{R}^d$ such that:

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Conceptual Punchline:

- A Linear MDP assumes the **Transition Matrix** is low rank ($\text{rank} \leq d$).
- Bellman Rank assumes the **Error Matrix** is low rank ($\text{rank} \leq d$).
- We do not need to know the features X_h, W_h ; we only assume they exist.

Example: LinMDPs have Bellman Rank d

Why did today's Linear MDP model work? Let's check its Bellman Rank:

- **LinMDP Hypothesis:** $Q_{h,g}(s, a) = \phi(s, a)^\top w_{h,g}$.
- **Bellman Error:** $\mathcal{E}_h^Q(f, g) = \mathbb{E}_{d_h^{\pi_f}}[\phi(s, a)^\top w_{h,g} - r_h(s, a) - \mathbb{E}_{P^\star}[V_{h+1,g}(s')]]$.

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From our "Magic Property" Lemma: $r_h(s, a) + \mathbb{E}_{P^*}[V_{h+1,g}]$ is linear in $\phi(s, a)$. Let its coefficient be $w_{lookahead}^*$.

$$\begin{aligned}\mathcal{E}_h^Q(f, g) &= \mathbb{E}_{(s,a) \sim d_h^{\pi_f}} \left[\phi(s, a)^\top (w_{h,g} - w_{lookahead}^*) \right] \\ &= \left\langle \underbrace{\mathbb{E}_{d_h^{\pi_f}}[\phi(s, a)]}_{X_h(f)}, \underbrace{w_{h,g} - w_{lookahead}^*}_{W_h(g)} \right\rangle\end{aligned}$$

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Result: LinMDPs are just one specific instance of the "Model Zoo" unified by Bellman Rank.

Algorithmic Teaser: Optimistic Elimination

How do we learn if the error matrix is low-rank but the features are unknown?

The PAC-RL Strategy:

- 1 **Eliminate:** If a hypothesis g shows a large empirical Bellman error on any past roll-in data, throw it out.
- 2 **Optimism:** Pick the f_t that predicts the highest value among survivors.
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The Logic for Next Time: Since the error matrix has rank d , we can only "fail" to eliminate a sub-optimal f_t a maximum of d times before we have spanned the space of possible errors.

Next Lecture: Formalizing the "Simplicity" of RL and the PAC proof.